Frequency domain Analysis for LTI systems

- Magnitude & Phase responses
- Design of first order & second order systems
- Linear phase, Minimum phase systems

$\{x(n)\}$ input → [LTI $\{h(n)\}$] → output $\{y(n)\} = \{h(n)\} * \{x(n)\}$

$X(\omega)$ → $H(\omega)$ (frequency response) → $Y(\omega) = H(\omega) X(\omega)$

$\{h(n)\} \xleftrightarrow{DTFT} H(\omega)$; $-\pi \le \omega \le \pi$

$$H(\omega) = |H(\omega)| e^{j \angle H(\omega)}$$

$|H(\omega)| \Rightarrow$ Magnitude response

$\angle H(\omega) \Rightarrow$ Phase response.

Recall connection between frequency response & transfer function

$\{h(n)\} \xleftrightarrow{z} H(z)$ with ROC R (includes unit circle)

$$H(\omega) = H(z)\Big|_{@\ z = e^{j\omega}}$$

Magnitude Response

$H(z)$ Rational transfer function ROC include unit circle

(easy to implement)

(LCCDE)

$$H(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

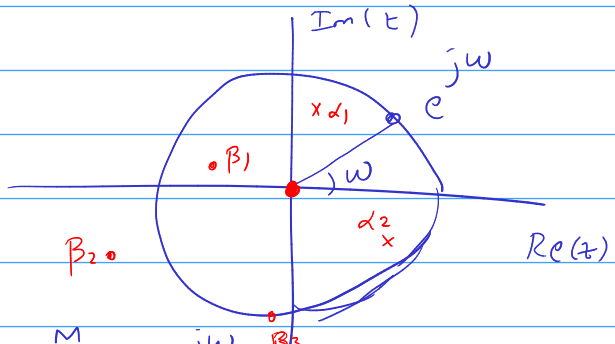
$$= b_0 \frac{\prod_{k=1}^{M} (1 - \beta_k z^{-1})}{\prod_{k=1}^{N} (1 - \alpha_k z^{-1})}$$

$\alpha_k \rightarrow$ poles $\beta_k \rightarrow$ zeros

$$H(z) = b_0 z^{(N-M)} \frac{\prod_{k=1}^{M} (z - \beta_k)}{\prod_{k=1}^{N} (z - \alpha_k)}$$

$$H(\omega) = H(z)\Big|_{z = e^{j\omega}}$$

$$H(\omega) = H(e^{j\omega})$$



$$H(\omega) = b_0 e^{j\omega(N-M)} \frac{\prod_{k=1}^{M} (e^{j\omega} - \beta_k)}{\prod_{k=1}^{N} (e^{j\omega} - \alpha_k)}$$

Magnitude response

$$|H(\omega)| = \underbrace{|b_0|}_{\text{const.}} \underbrace{|e^{j\omega(N-M)}|}_{1} \frac{\prod_{k=1}^{M} |e^{j\omega} - \beta_k|}{\prod_{k=1}^{N} |e^{j\omega} - \alpha_k|}$$

$|e^{j\omega} - \beta_k| \Rightarrow$ distance between points $e^{j\omega}$ & β_k in complex plane.

Locations of poles & zeros shape the magnitude response.

Remarks

1. If e^{jw} is closer to a zero $H(w)$ will be small for that w.

2. If e^{jw} is closer to a pole $H(w)$ will be large

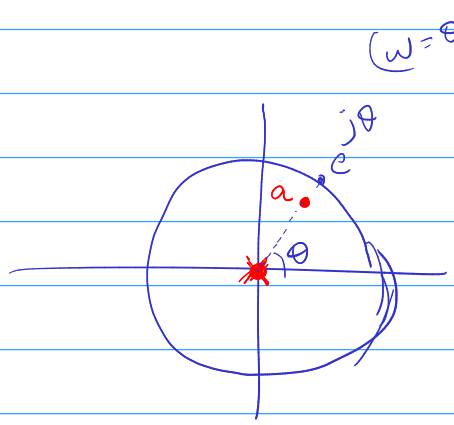
3. Poles or Zeros at origin does not affect magnitude response

4. If there is a zero on unit circle $e^{jw_0} = \beta_k$ then $H(w_0) = 0$

First Order System

1 Single complex zero

$$H(z) = 1 - a z^{-1}$$



$$h(n) = \{ \underset{\downarrow}{1}, -a \}$$

$$H(\omega) = 1 - a e^{-j\omega}$$

$$a = r e^{j\theta}$$

$$H(\omega) = 1 - r e^{-j(\omega-\theta)}$$

$$|H(\omega)|^2 = H(\omega)\, H^*(\omega)$$

$$= \left(1 - r e^{-j(\omega-\theta)}\right)\left(1 - r e^{+j(\omega-\theta)}\right)$$

$$= 1 + r^2 - 2r\cos(\omega-\theta)$$

$|H(\omega)|^2$ will reach minimum value at $\omega = \theta$

$$|H(\omega=\theta)|^2 = (1-r)^2$$

$|H(\omega)|^2$ will reach maximum value $\omega = \pi + \theta$

$$|H(\omega=\pi+\theta)|^2 = (1+r)^2$$

In dB Scale

$$10 \log |H(\omega)|^2 = 20 \log |H(\omega)| \quad dB$$

$$= 10 \log (1 + r^2 - 2r \cos(\omega - \theta))$$

